PROJECT-BASED ASSESSMENT

NATURAL LANGUAGE PROCESSING (NLP) PROJECT

Course Code: CSE472

Student Name(s): Sudhanshu Singh

Roll Number(s): 29

Registration Number: 12405898

Project Title: FinSight – Financial News & Market Sentiment Analyzer

Q1. NLP Task / Application

☑ Sentiment Analysis

Q2. Problem Statement and Objective

Financial news significantly influences stock market movements, but manually analyzing large volumes of financial news is time-consuming. FinSight aims to automate financial news sentiment analysis by classifying news into positive, negative, and neutral categories, helping users understand market sentiment for specific companies and stocks.

Q3. Is a Working Implementation Available?

☑ Jupyter/Colab Notebook

☑ Web Application

Demo / Application Link:
https://finsight-12405898.streamlit.app/

Q4. Data Preprocessing Steps Used

☑ Lowercasing

☑ Tokenization

☑ Stop-word Removal

☑ Stemming

☑ Lemmatization

☑ Punctuation / Special Character Removal

☑ Handling Missing or Duplicate Data

Q5. Text Representation / Feature Extraction Used

☑ TF-IDF

☑ Word2Vec / GloVe

☑ Tokenizer / Transformer Embeddings

Q6. Algorithm / Model Used

☑ Naive Bayes / Logistic Regression / SVM — Logistic Regression and SVM

☑ RNN / LSTM / GRU / BiLSTM — BiLSTM

☑ Transformer — BERT / FinBERT

Exact Model Names and Important Configuration:

1. Logistic Regression (trained using TF-IDF features)
2. Support Vector Machine (SVM) (trained using TF-IDF features)
3. Bidirectional Long Short-Term Memory (BiLSTM)
4. FinBERT (financial-domain transformer model)

Q7. Why Were These Models Selected?

The models were selected to compare traditional machine learning approaches with deep learning and transformer-based approaches for financial sentiment analysis. Logistic Regression and SVM were used to establish baseline performance. BiLSTM was selected to

capture contextual information in sequential text, while FinBERT was selected because it is specifically designed for understanding financial language and sentiment.

Q8. Tools, Libraries, and Platform Used

Programming Language: Python

Platforms: Google Colab, VS Code, Streamlit Cloud

Libraries / Frameworks:

- Pandas
- NumPy
- Scikit-learn
- NLTK
- TensorFlow / Keras
- PyTorch
- Hugging Face Transformers
- Streamlit
- Requests
- YFinance

Q9. Evaluation Metrics Used

☑ Accuracy

☑ Precision

☑ Recall

☑ F1-score

☑ Confusion Matrix

Q10. Main Results Obtained

The project evaluated four models on a test dataset containing 1,065 samples.

Model	Accuracy	Weighted F1-score
Logistic Regression	73%	0.74
SVM	74%	0.74
BiLSTM	72%	0.72
FinBERT	79%	0.79

FinBERT achieved the best overall performance with 79% accuracy and a weighted F1-score of 0.79, outperforming the traditional machine learning and BiLSTM models.

Q11. Major Limitations or Errors Observed

1. Negative sentiment classification was comparatively weak.
2. Some financial headlines were misclassified due to ambiguous wording.
3. Traditional machine learning models struggled with contextual understanding.
4. BiLSTM performed worse than FinBERT.
5. Class imbalance affected model performance.

Q12. Application / Demo Link

FinSight – Financial News & Market Sentiment Analyzer

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